

Microelectronics

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Contents

1	Microelectronics	1
I	Signal	1
i	Signal	1
ii	Signal source	1
iii	Transducer	1
iv	Signal processing	1
v	Symbol convention	1
II	Amplifier (amp)	1
i	Amplifier	1
ii	Diagram	2
iii	Two-port model	2
iv	Circuit model of voltage amplifier.	2
v	Circuit model of current amplifier	2
vi	Circuit model of transconductance amplifier.	3
vii	Circuit model of transresistance amplifier.	3
viii	Amplifier gain in decibels (dB).	4
ix	Amplifiers in cascade	4
x	Saturation.	4
xi	S parameter	4
xii	Unilateral amplifier	4
xiii	Power supply	4
III	Operational Amplifier (op amp or op-amp)	5
i	Definition	5
ii	Ideal op amp	5
iii	General amplifier with inverting configuration.	5
iv	Inverting amplifier	5
v	General amplifier with noninverting configuration	6
vi	Noninverting amplifier.	6
vii	Unity gain amplifier, unity gain buffer, or voltage follower	6
viii	Summing amplifier	6

ix Difference amplifier or differential amplifier	7
x Comparator	7
xi Inverting integrator	7
xii Inverting differentiator	7
xiii Active filter	8

1 Microelectronics

I Signal

i Signal

A signal is a function that conveys information about a phenomenon. Signal processing is usually most conveniently performed by electronic systems on voltage or current signals, especially the former. Only voltage and current signals are studied here.

ii Signal source

A signal source is typically represented in the Thévenin form, a voltage source in a series connection with a resistance, or the Norton form, a current source in a parallel connection with a resistance.

iii Transducer

A transducer is a device that convert a signal to a voltage or current signal or vice versa.

iv Signal processing

Signal processing includes analog signal processing (ASP), analog-to-digital converter (ADC), digital signal processor (DSP), digital-to-analog converter (DAC), amplifier, etc.

v Symbol convention

- Lowercase character with uppercase subscript means total signal, e.g. $i_C(t)$.
- Lowercase character with lowercase subscript means AC signal, e.g. $i_c(t)$.
- Uppercase character with uppercase subscript means DC signal, e.g. I_C .
- Uppercase character with lowercase subscript means peak signal, e.g. I_c .

Their relation is

$$i_C(t) = i_c(t) + I_C$$

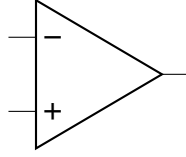
$$i_c(t) = I_c \sin(\omega t + \varphi)$$

II Amplifier (amp)

i Amplifier

An electronic device which amplifies an electrical signal by a scale called amplifier gain A and optionally convert voltage signal to current signal or vice versa.

ii Diagram



iii Two-port model

An amplifier can be modeled as a two port model, where an one input port and one output port is related by the amplifier, which acts as a controlled source controlled by the input port voltage or current which enforces voltage or current on the output port.

iv Circuit model of voltage amplifier

A voltage amplifier draws current from a source v_s with internal resistance R_s with input resistance R_i and voltage v_{in} across it and supplies $v_c = \mu v_{in}$ to a load network with output resistance R_o and load R_L , where μ , also denoted A_{vo} , is open-circuit voltage gain factor.

$$\frac{v_{in}}{v_s} = \frac{R_i}{R_s + R_i}$$

$$\frac{v_c}{v_{in}} = \mu$$

$$\frac{v_{out}}{v_c} = \frac{R_L}{R_o + R_L}$$

Voltage gain:

$$A_v = \frac{v_{out}}{v_{in}} = \mu \frac{R_i R_L}{(R_s + R_i)(R_o + R_L)}$$

$|A| < |\mu|$ due to loading effects.

A good amplifier requires small loading effect:

$$R_i \gg R_s, \quad R_o \ll R_L, \quad A_v \approx \mu.$$

v Circuit model of current amplifier

A current amplifier draws current from a current source i_s with internal resistance R_s in parallel with input resistance R_i and current i_{in} through it and supplies $i_c = \beta i_{in}$ by CCCS to a load network with output resistance R_o in parallel and load R_L , where β , also denoted A_{cs} , is short-circuit current gain factor.

$$\frac{i_{in}}{i_s} = \frac{R_s}{R_s + R_i}$$

$$\frac{i_c}{i_{in}} = \beta$$

$$\frac{i_{out}}{i_c} = \frac{R_o}{R_o + R_L}$$

Current gain:

$$A_c = \frac{i_{out}}{i_{in}} = \beta \frac{R_s R_o}{(R_s + R_i)(R_o + R_L)}$$

$|A| < |\beta|$ due to loading effects.

A good amplifier requires small loading effect:

$$R_i \ll R_s, \quad R_o \gg R_L, \quad A_c \approx \beta.$$

vi Circuit model of transconductance amplifier

A transconductance amplifier draws current from a voltage source v_s with internal resistance R_s in series with input resistance R_i and current i_{in} through it and supplies $i_c = g v_{in}$ by VCCS to a load network with output resistance R_o in parallel and load R_L , where μ is open-circuit transconductance factor.

$$\begin{aligned} \frac{v_{in}}{v_s} &= \frac{R_i}{R_s + R_i} \\ \frac{i_c}{v_{in}} &= g \\ \frac{i_{out}}{i_c} &= \frac{R_o}{R_o + R_L} \end{aligned}$$

Gain:

$$A_g = \frac{i_{out}}{v_{in}} = g \frac{R_i R_o}{(R_s + R_i)(R_o + R_L)}$$

$|A| < |g|$ due to loading effects.

A good amplifier requires small loading effect:

$$R_i \gg R_s, \quad R_o \gg R_L, \quad A_g \approx g.$$

vii Circuit model of transresistance amplifier

A transresistance amplifier draws current from a current source i_s with internal resistance R_s in parallel with input resistance R_i and current i_{in} through it and supplies $v_c = r i_{in}$ by CCVS to a load network with output resistance R_o in series and load R_L , where r is short-circuit transresistance factor.

$$\begin{aligned} \frac{i_{in}}{i_s} &= \frac{R_s}{R_s + R_i} \\ \frac{v_c}{i_{in}} &= r \\ \frac{v_{out}}{v_c} &= \frac{R_L}{R_o + R_L} \end{aligned}$$

Gain:

$$A_r = \frac{v_{out}}{i_{in}} = r \frac{R_s R_L}{(R_s + R_i)(R_o + R_L)}$$

$|A| < |r|$ due to loading effects.

A good amplifier requires small loading effect:

$$R_i \ll R_s, \quad R_o \ll R_L, \quad A_c \approx r.$$

viii Amplifier gain in decibels (dB)

A power gain A_p in dB is defined as

$$G_p = 10 \log_{10}(A_p).$$

A voltage gain A_v in dB is defined as

$$G_v = 20 \log_{10}(A_v).$$

A current gain A_c in dB is defined as

$$G_c = 20 \log_{10}(A_c).$$

ix Amplifiers in cascade

Output of an amplifier as input of another amplifier.

x Saturation

The linear characteristic only remains within a range. A positive and negative saturation levels can be observed.

xi S parameter

If the energy enters through port i and emerges from port j , the S parameter is named S_{ji} .

xii Unilateral amplifier

An unilateral amplifier is an amplifier in which the signal transfer is effectively one-way, that is, the output signal is only dependent on the input signal but not what is connected to the output terminal. That is for each output port j and each input port i , $S_{ij} = 0$.

xiii Power supply

To amplify signal, power supplies are needed, but often omitted in the diagram.

The DC power delivered is the sum of power from the positive and negative power supply terminals V_{CC} and V_{EE}

$$P_{DC} = V_{CC}I_{CC} + V_{EE}I_{EE}$$

Suppose input and output signals have power P_I and P_L .

$$P_{DC} + P_I = P_L + P_{\text{dissipated}}$$

where $P_{\text{dissipated}}$ is the power dissipated.

III Operational Amplifier (op amp or op-amp)

i Definition

An op amp has five terminals:

- v_+ : non-inverting input voltage of the terminal at the + sign;
- v_- : inverting input voltage of the terminal at the – sign;
- v_o : output voltage of the terminal at the opposite side of inputs;
- V_{CC+} or V_{CC} : positive supply voltage, denoted on the + side of an op-amp or omitted;
- V_{CC-} or V_{EE} : negative supply voltage, denoted on the – side of an op-amp or omitted.

$$v_o = A(v_+ - v_- - v_{os}),$$

where A is open-loop gain, and v_{os} is input offset voltage.

ii Ideal op amp

- Infinite open-loop gain, infinite bandwidth, and zero output impedance: $A = \infty$ and is independent of frequency and output voltage.
- Zero input offset voltage: $v_{os} = 0$.
- No current flow into either of the input terminals.

This is a linear element with inputs v_+ and v_- and output v_o .

iii General amplifier with inverting configuration

A general amplifier with inverting configuration consists of an op amp where input voltage v_i is connected to v_- through an input impedance Z_i , v_+ is connected to ground, and v_o is fed back to v_- through a feedback impedance Z_f .

$$I_i = \frac{V_i}{Z_i} = I_f = \frac{-V_o}{Z_f},$$

$$V_o = -\frac{Z_f}{Z_i} V_i.$$

When $\left| \Re\left(\frac{Z_f}{Z_i}\right) \right| < 1$, it's called an attenuator.

iv Inverting amplifier

An inverting amplifier consists of an op amp where input voltage v_i is connected to v_- through an input resistor R_i , v_+ is connected to ground, and v_o is fed back to v_- through a feedback resistor R_f .

$$i_i = \frac{v_i}{R_i} = i_f = \frac{-v_o}{R_f},$$

$$v_o = -\frac{R_f}{R_i} v_i.$$

v General amplifier with noninverting configuration

A general amplifier with noninverting configuration consist of an op amp where input voltage v_i is connected to v_+ , v_- is connected ground through a ground impedance Z_g , and v_o is fed back to v_- through a feedback impedance Z_f .

$$V_i = V_o \frac{Z_g}{Z_f + Z_g},$$
$$V_o = \left(1 + \frac{Z_f}{Z_g}\right) V_i.$$

vi Noninverting amplifier

A noninverting amplifier consist of an op amp where input voltage v_i is connected to v_+ , v_- is connected ground through a ground resistor R_g , and v_o is fed back to v_- through a feedback resistor R_f .

$$v_i = v_o \frac{R_g}{R_f + R_g},$$
$$v_o = \left(1 + \frac{R_f}{R_g}\right) v_i.$$

vii Unity gain amplifier, unity gain buffer, or voltage follower

A unity gain amplifier consist of an op amp where input voltage v_i is connected to v_+ and v_o is fed back to v_- .

$$v_o = v_i.$$

viii Summing amplifier

A summing amplifier consists of an op amp, input voltages $v_1, v_2, \dots, v_n$, n input resistors $R_1, R_2, \dots, R_n$, and a feedback resistor R_f , where for any $k \in \mathbb{N} \wedge k \leq n$, v_k is connected to v_- through R_k , v_+ is connected to ground, and v_o is fed back to v_- through R_f .

$$i_k = \frac{v_k}{R_k},$$
$$\sum_{k=1}^n i_k = i_f = \frac{-v_o}{R_f},$$
$$v_o = -R_f \sum_{k=1}^n \frac{v_k}{R_k}.$$

When $R_k = R$ for all $k \in \mathbb{N} \wedge k \leq n$:

$$v_o = -\frac{R_f}{R} \sum_{k=1}^n v_k.$$

ix Difference amplifier or differential amplifier

A difference amplifier consists of a minuend v_1 , a subtrahend v_2 , and four resistors R_1, R_2, R_3, R_4 satisfying the condition

$$\frac{R_1}{R_2} = \frac{R_3}{R_4},$$

where v_1, v_2 are connected to v_-, v_+ through R_1, R_2 respectively, v_o is fed back to v_- through R_3 , and v_+ is connected to ground through R_4 .

$$\begin{aligned} v_+ &= v_2 \frac{R_4}{R_2 + R_4}, \\ \frac{v_1 - v_-}{R_1} &= \frac{v_- - v_o}{R_3}, \\ \frac{v_1}{R_2} - v_2 \frac{R_4}{R_2 + R_4} \frac{R_2 + R_4}{R_2 R_4} + \frac{v_o}{R_4} &= 0, \\ v_o &= \frac{R_4}{R_2} (v_2 - v_1). \end{aligned}$$

When $R_1 = R_2 = R_3 = R_4$:

$$v_o = (v_2 - v_1).$$

x Comparator

A comparator is an op amp with v_+ and v_- connected to the voltage sources to be compared, in which v_o is high when $v_+ > v_-$ and low when $v_+ < v_-$.

xi Inverting integrator

An inverting integrator consists of an op amp where input voltage v_i is connected to v_- through an input resistor R_i , v_+ is connected to ground, and v_o is fed back to v_- through a feedback capacitor C_f .

Time domain:

$$\begin{aligned} i_i &= \frac{v_i}{R_i} = i_f = -C_f \frac{dv_o}{dt}, \\ v_o &= v_o(0^-) - \int_0^t \frac{v_i(\tau)}{R_i C_f} d\tau. \end{aligned}$$

Laplace domain:

$$\begin{aligned} I_i &= \frac{V_i}{R_i} = I_f = -C_f s V_o + C_f v_o(0^-), \\ V_o &= \frac{v_o(0^-)}{s} - \frac{V_i}{R_i C_f s}. \end{aligned}$$

xii Inverting differentiator

An inverting differentiator consists of an op amp where input voltage v_i is connected to v_- through an input capacitor C_i , v_+ is connected to ground, and v_o is fed back to v_- through a feedback resistor R_f .

Time domain:

$$i_i = C_i \frac{dv_i}{dt} = i_f = \frac{-v_o}{R_f},$$
$$v_o = R_f C_i \left(v_i(0^-) - \int_0^t v_i(\tau) d\tau \right).$$

Laplace domain:

$$I_i = C_i s V_i - C_i v_i(0^-) = I_f = \frac{-V_o}{R_f},$$
$$V_o = R_f C_i (v_i(0^-) - s V_i).$$

xiii Active filter

Given a (passive or active) filter v_f of an input v_s and an inverting amplifier, noninverting amplifier, or unit gain buffer enforcing $v_o = K v_i$, by connecting the ground and v_i of the amplifier to the two terminals of the v_f , the two terminals ground and v_o become an active filter of v_s enforcing $v_o = K v_f$.